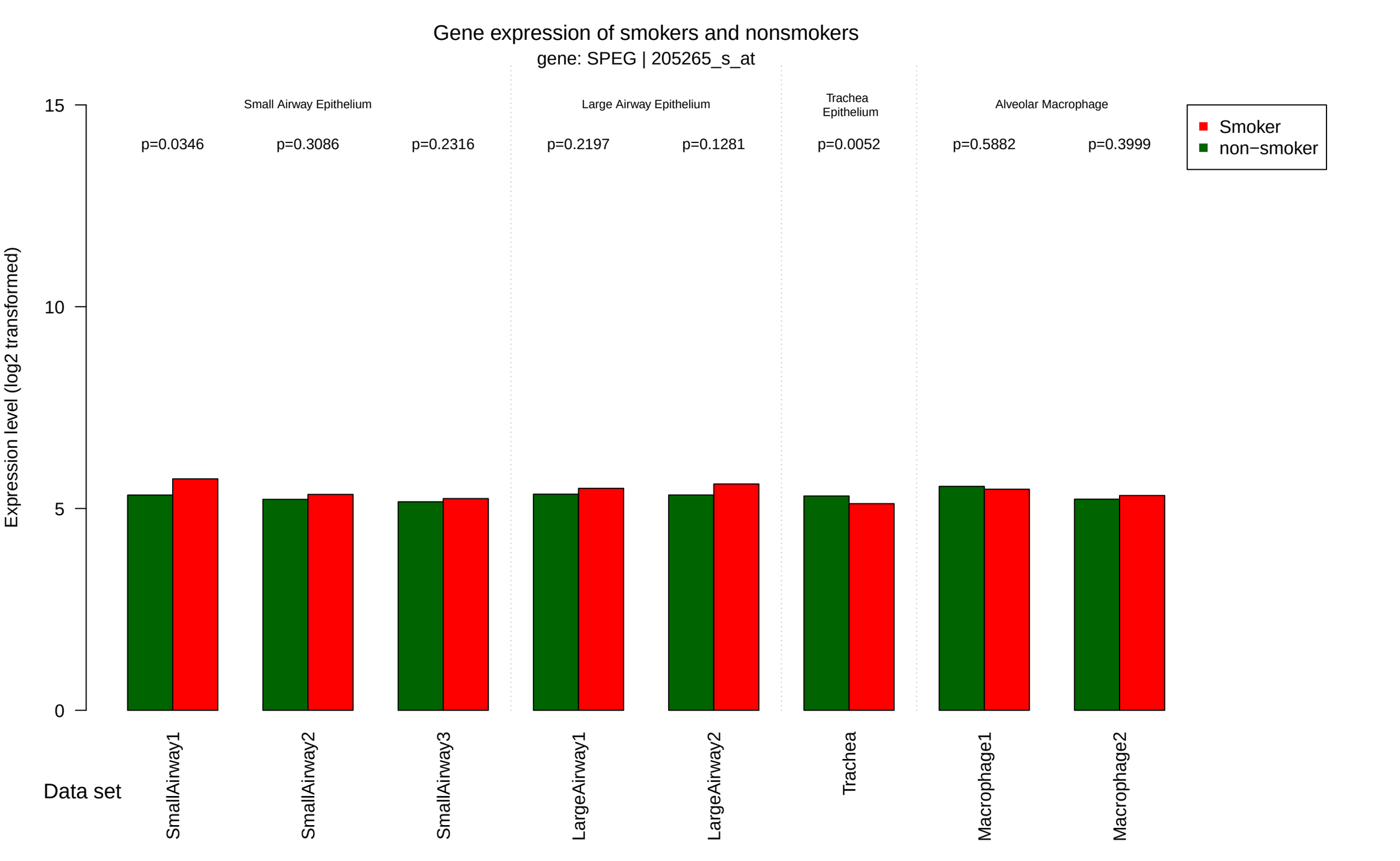
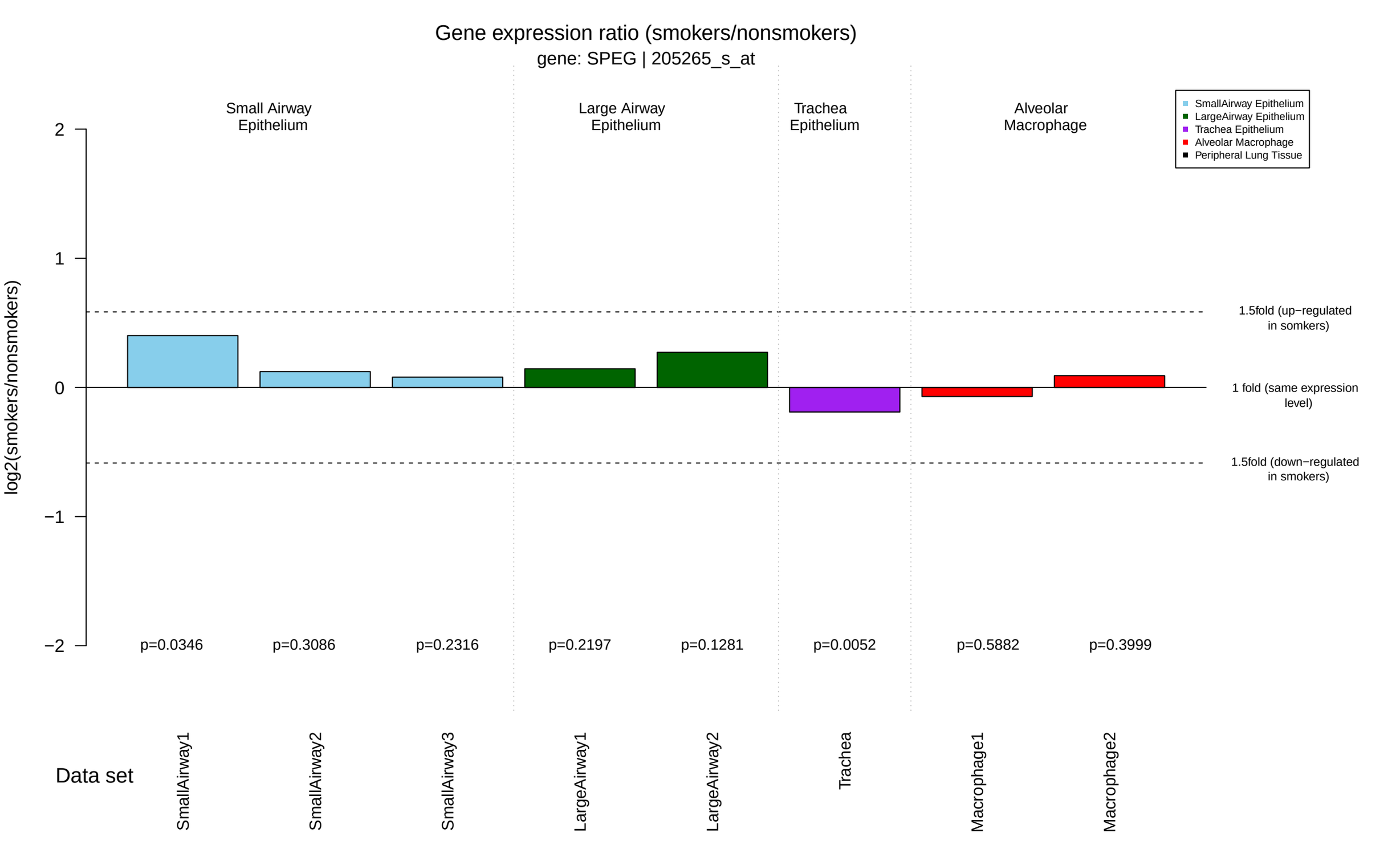
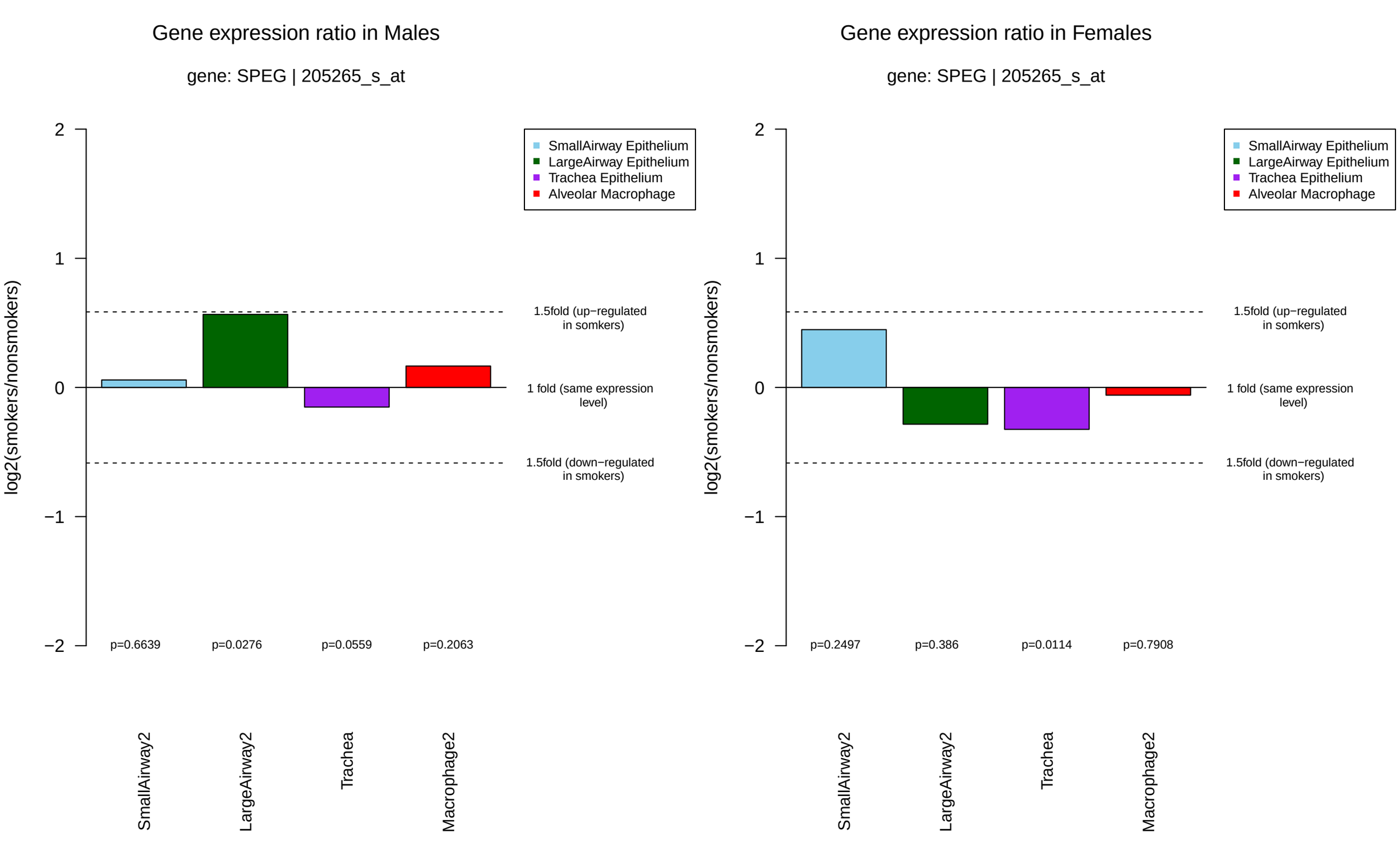
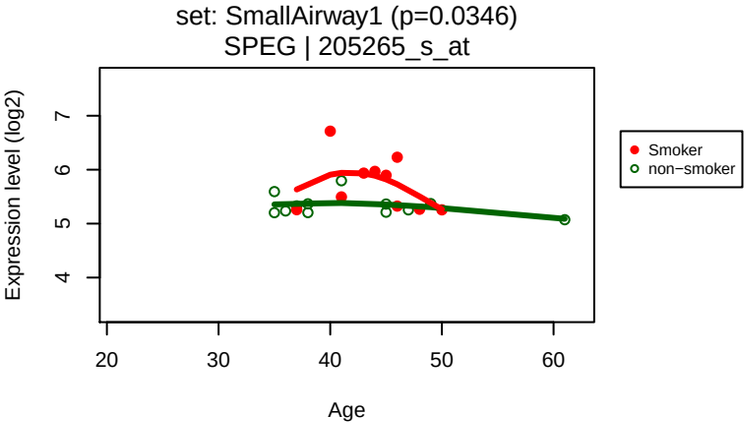
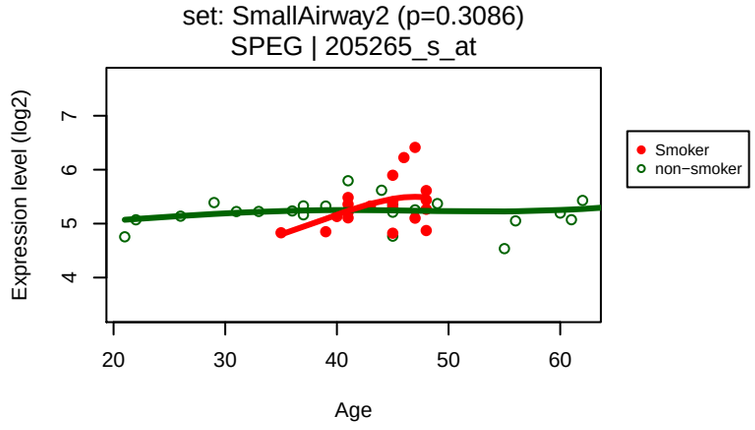
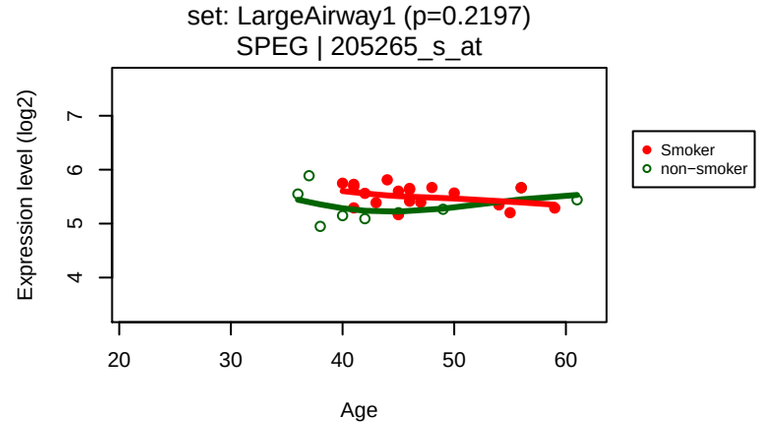

Gene expression of smokers and nonsmokers
gene: SPEG | 205265_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0346
p=0.3086
p=0.2316
p=0.2197
p=0.1281
p=0.0052
p=0.5882
p=0.3999
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

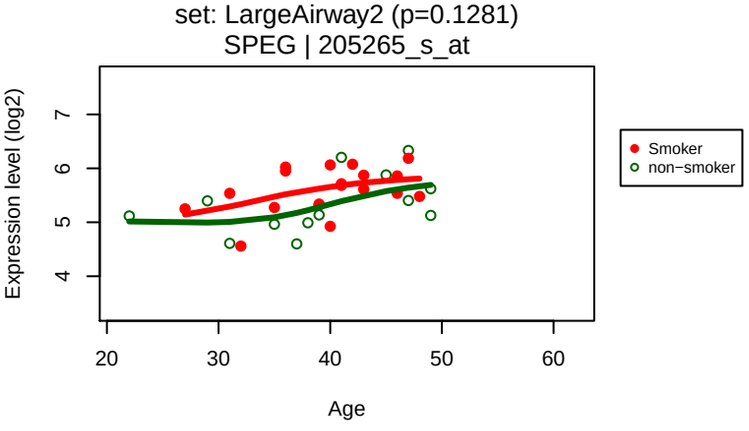

Gene expression ratio (smokers/nonsmokers)
gene: SPEG | 205265_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0346
p=0.3086
p=0.2316
p=0.2197
p=0.1281
p=0.0052
p=0.5882
p=0.3999
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

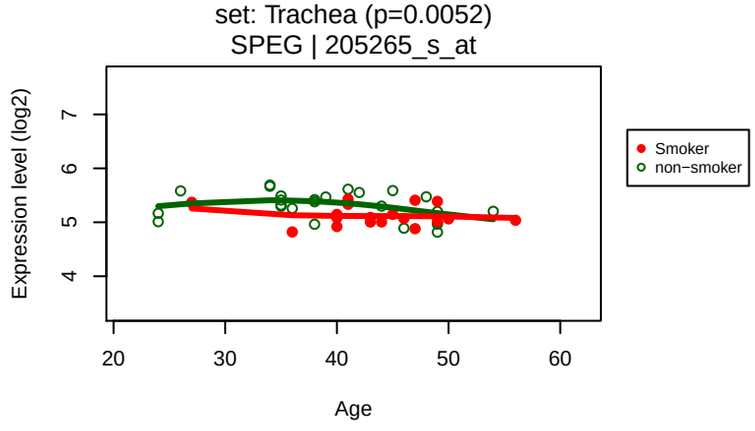

Gene expression ratio in Males
gene: SPEG | 205265_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6639
p=0.0276
p=0.0559
p=0.2063
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: SPEG | 205265_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2497
p=0.386
p=0.0114
p=0.7908
SmallAirway2
LargeAirway2
Trachea
Macrophage2

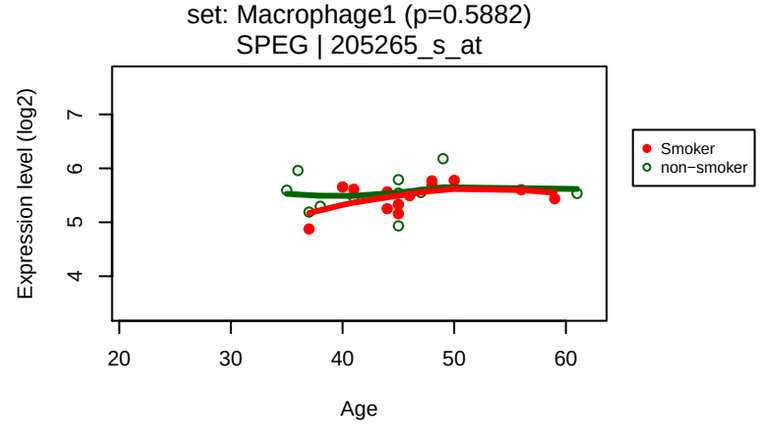

set: SmallAirway1 (p=0.0346)
SPEG | 205265_s_at
Expression level (log2)
Age
Smoker
non−smoker

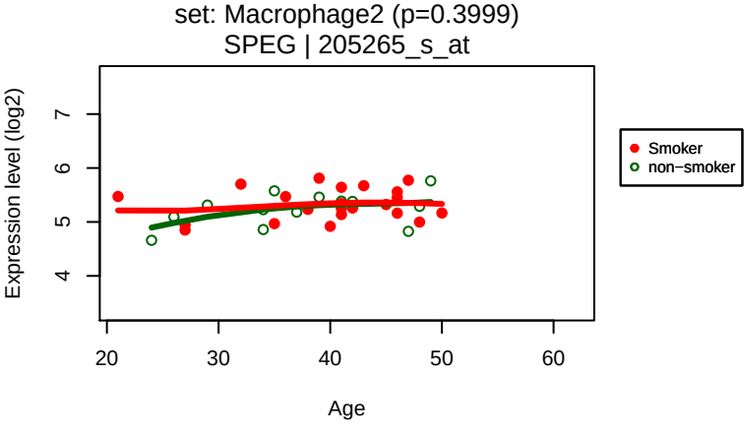

set: SmallAirway2 (p=0.3086)
SPEG | 205265_s_at
Expression level (log2)
Age
Smoker
non−smoker

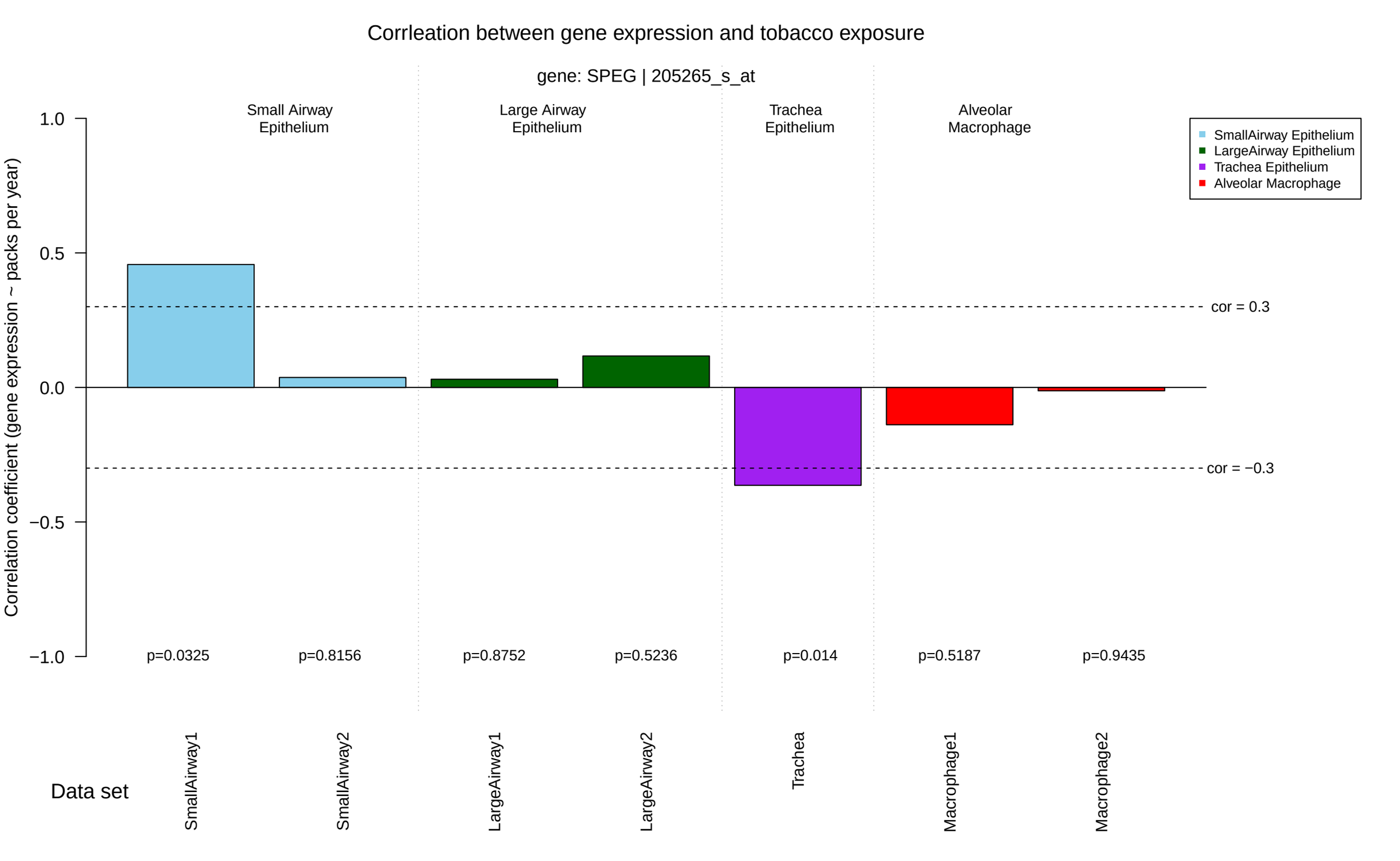

set: LargeAirway1 (p=0.2197)
SPEG | 205265_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.1281)
SPEG | 205265_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0052)
SPEG | 205265_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.5882)
SPEG | 205265_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.3999)
SPEG | 205265_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: SPEG | 205265_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0325
p=0.8156
p=0.8752
p=0.5236
p=0.014
p=0.5187
p=0.9435
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2